

SYDNEY BOYS' HIGH SCHOOL

MOORE PARK, SURRY HILLS



Year 11 YEARLY EXAMINATIONS – August 2000

MATHEMATICS

Time allowed — Two Hours
Examiners: E.Choy, A.M.Gainford

DIRECTIONS TO CANDIDATES

- *ALL* questions may be attempted.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- Use a new booklet for each question.
- If required, additional booklets may be obtained from the Examination Supervisor upon request.

Question 1. (18 Marks) (Start a new booklet.)

- (a) Calculate $\sqrt{\frac{67}{4 \cdot 7 \times 2 \cdot 3}}$ correct to two decimal places.
- (b) Simplify $x - 2(3 - x)$.
- (c) Solve the equation $\frac{x}{3} - \frac{x+1}{2} = 4$.
- (d) Simplify $\sqrt{32} - \sqrt{8}$.
- (e) Find x if $\log_3 x = 4$.
- (f) Find θ to the nearest minute if $0^\circ \leq \theta \leq 90^\circ$ and $\cos \theta = 0.613$.
- (g) Solve the equation $3x^2 = 12$.
- (h) Graph on a number line the solution of the inequality $|x - 2| < 3$.
- (i) Simplify $\frac{(xy^2)^3}{x^3y^2}$.
- (j) Find the exact value of $\sin 135^\circ + \tan 480^\circ$.
Express your answer as a single fraction with rational denominator.
- (k) Given that $f(x) = x - \frac{1}{x}$:
- (i) Find $f(4)$.
- (ii) Show that $f(x)$ is an odd function.

Question 2. (18 Marks) (Start a new booklet.)

(a) Simplify $\frac{x^2 - y^2}{(x + y)^2}$.

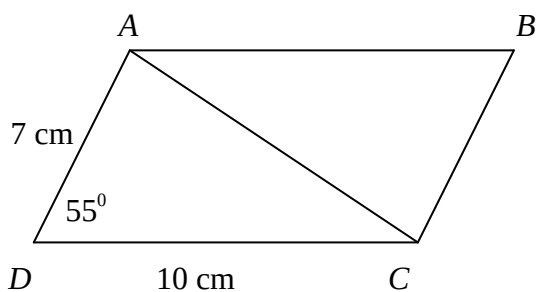
(b) Sketch the graphs of the following:

(i) $y = (x - 1)^2$

(ii) $y = \sqrt[3]{x}$

(c) Express the recurring decimal $0.4232323\dots$ as a common fraction.

(d) Given the parallelogram $ABCD$:

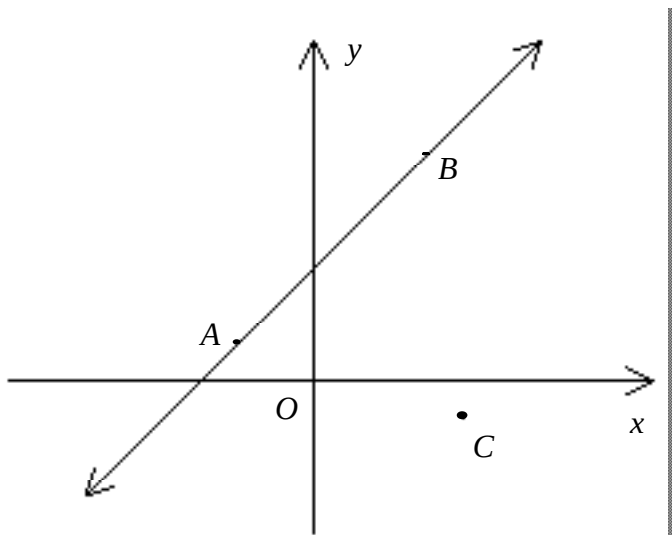


(i) Find the length of the diagonal AC , correct to 2 decimal places.

(ii) Calculate the area of the parallelogram, correct to 2 decimal places.

(e) State the natural domain and range of the function $f(x) = \sqrt{9 - x^2}$.

(f)



The diagram above shows the points $A(-2,1)$, $B(3,5)$, and $C(4, -1)$.

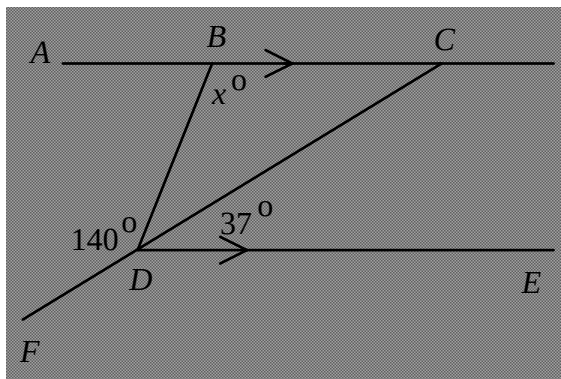
Copy the diagram to your answer booklet.

- (i) Find the equation of the line through the points A and B .
- (ii) Write the equation of the line through C perpendicular to AB .
- (iii) Hence or otherwise find the distance from C to AB .

Question 3. (18 Marks) (Start a new booklet.)

- (a) (i) Find the points of intersection of the line $y = 4 - x$ and the circle $x^2 + y^2 = 16$.
- (ii) Hence sketch the region where $y \geq 4 - x$ and $x^2 + y^2 < 16$ hold simultaneously.

(b)

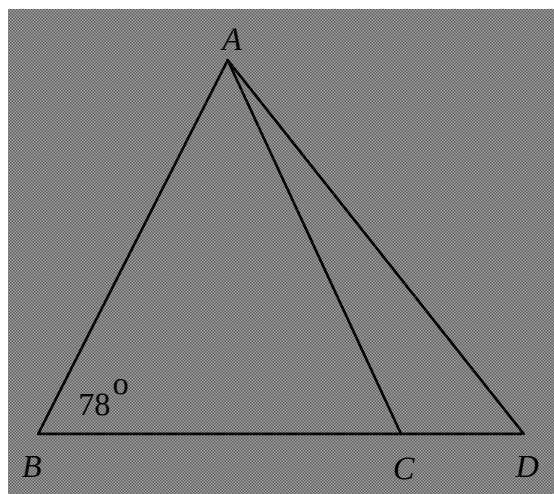


$AC \parallel DE$, CDF is a straight line.

Find the measure of x .

- (c) $AB = AC = BD$.

Determine the size of $\angle ADB$ and $\angle DAC$ giving reasons.



- (d) Solve the following equations:

(i) $x^4 - 13x^2 - 9 = 0$

(ii) $9^x - 8(3^x) - 9 = 0$

- (e) For the parabola $y^2 - 6y - 2x + 7 = 0$ write down the

- (i) equation of the axis of symmetry
- (ii) coordinates of the vertex
- (iii) equation of the directrix and coordinates of the focus.

- (f) Show that the expression $x^2 - (k + 2)x + (3k + 6)$ is positive definite if $-2 < k < 10$.

Question 4. (18 Marks) (Start a new booklet.)

(a) Differentiate the following with respect to x :

(i) $-x^3 + 2x^2 + \frac{1}{2}$

(ii) $\sqrt{7x}$

(iii) $\frac{ax^3 - bx^2 + cx}{x^2}$

(b) (i) Use the product rule to find $\frac{dy}{dx}$ if $y = 2x(x+1)^8$.

(ii) Differentiate $y = \frac{5+t}{5-2t}$ by using the quotient rule.

(iii) If $f(x) = 3x + \frac{1}{x^3}$, find

(α) $f'(2)$

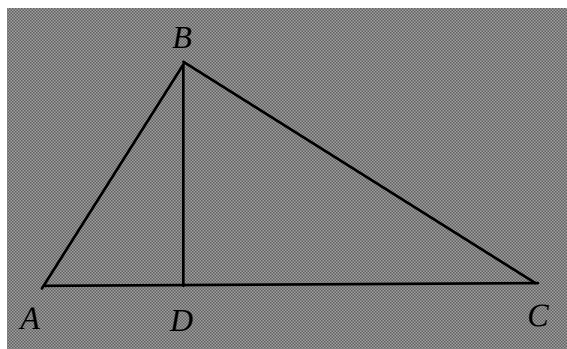
(β) $f''(2)$

(c) (i) For the curve $y = \frac{1}{x^2}$, find the gradient of the tangent to the curve at the point $\left(2, \frac{1}{4}\right)$. Also find the gradient of the normal to the curve at this point.

(ii) Given that $f(x)$ is defined as below, find the value of $f(-3) + f(4) + f(-1)$.

$$f(x) = \begin{cases} -5 & \text{for } x \leq -3 \\ 2x & \text{for } -3 < x < 0 \\ x^2 & \text{for } x \geq 0 \end{cases}$$

(d)



In the diagram ABC and ABD are right-angled triangles. $\angle ADB = \angle ABC = 90^\circ$.

Copy the diagram to your answer booklet.

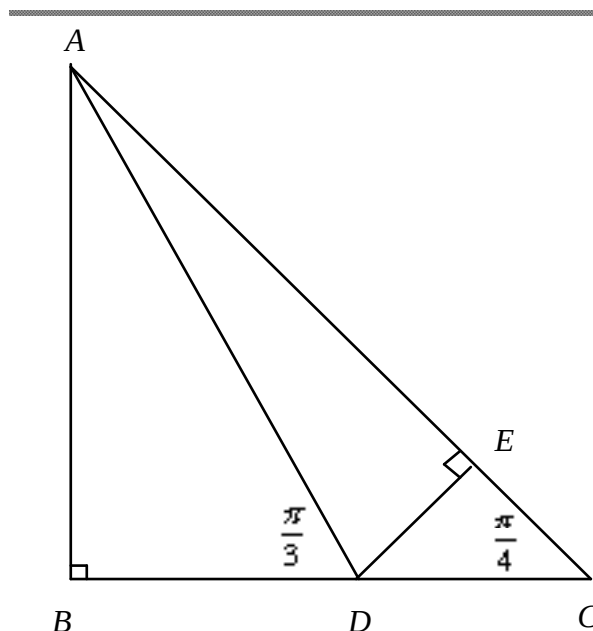
- (i) Prove that $\triangle ABD \sim \triangle ACB$.
 - (ii) Hence find AB if $AD = 4$ cm and $DC = 5$ cm.
- (e) Find the value or values of k that will make the equation $x^2 + 16x - 4k = 0$ have:
- (i) equal roots
 - (ii) two distinct real roots
 - (iii) roots which are reciprocals of one another
 - (iv) the sum of roots equal to their product.

Question 5. (18 Marks) (Start a new booklet.)

- (a) The point $P(x,y)$ moves in the plane so that its distance from a point $A(-2,4)$ is always twice its distance from the point $B(4,1)$.
- (i) Write down an expression connecting PA and PB .
- (ii) Hence find the locus of P .
- (b) Consider the function $y = \frac{1}{|x-1|}$.
- (i) What is the natural domain of the function?
- (ii) Write down the equations of the two branches of the function, and sketch its graph.
- (c) Solve the equation $8\cos^2 x^\circ = 2\sin x^\circ + 7$ where $0^\circ \leq x^\circ \leq 360^\circ$. Give your answer correct to the nearest minute.
- (d) Prove the identity

$$\frac{1}{1+\sin\theta} + \frac{1}{1-\sin\theta} = 2\sec^2\theta$$

- (e) The diagram shows a right-angled triangle ABC , whose angle ACB is $\frac{\pi}{4}$. Line AD meets BC at D such that angle ADB is $\frac{\pi}{3}$, and length BD is one unit. Line DE meets line AC at right angles.



Find in **exact** form the:

- (i) length AB
- (ii) length DC
- (iii) length DE
- (iv) $\angle DAC$ in terms of π , and

hence show that $\sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$.

This is the end of the paper.



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AUGUST 2000

YEARLY EXAMINATION

YEAR 11

Mathematics

Sample Solutions

Question 1

(a) $2 \cdot 49$
 (b) $x - 6 + 2x = 3x - 6$
 (c) $2x - (3x + 3) = 24$

$$-x - 3 = 24$$

$$\therefore x = -27$$

(d) $4\sqrt{2} - 2\sqrt{2} = 2\sqrt{2}$

(e) $x = 3^4 = 81$

(f) $\theta = \cos^{-1}(0.613) = 52^\circ 12'$

(g) $x^2 = 4$

$$\therefore x = \pm 2$$

(h) $-3 < x - 2 < 3$

$$\therefore -1 < x < 5$$



(i) $\frac{x^3 y^6}{x^3 y^2} = y^4$

(j) $\frac{1}{\sqrt{2}} - \sqrt{3} = \frac{1 - \sqrt{6}}{\sqrt{2}} = \frac{\sqrt{2} - \sqrt{12}}{2} = \frac{\sqrt{2} - 2\sqrt{3}}{2}$

(k) (i) $f(4) = 4 - \frac{1}{4} = 3\frac{3}{4}$

(ii) $f(-x) = (-x) - \frac{1}{(-x)}$

$$= -x + \frac{1}{x}$$

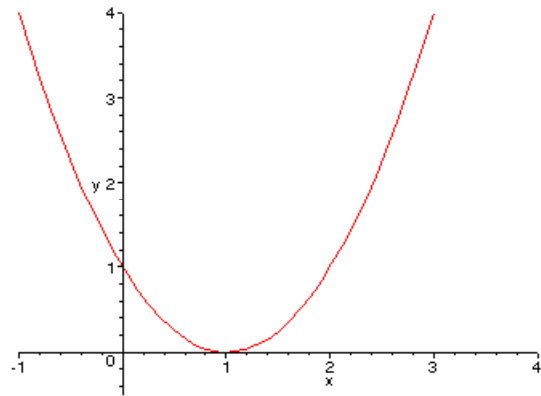
$$= -\left(x - \frac{1}{x}\right)$$

$$= -f(x)$$

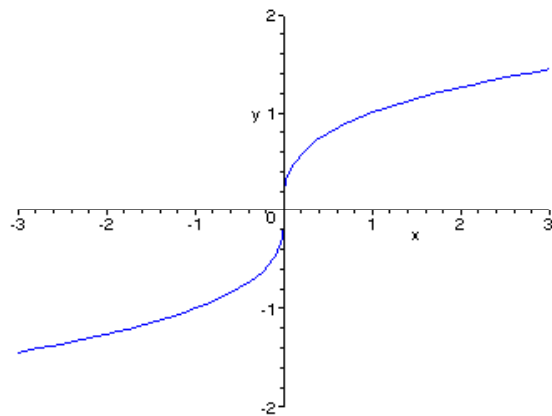
Question 2

(a)
$$\frac{(x+y)(x-y)}{(x+y)^2} = \frac{x-y}{x+y}$$

(b) (i)



(ii)



(c)
$$0.4\dot{2}\dot{3} = \frac{423-4}{990} = \frac{419}{990}$$

(d) (i)
$$AC^2 = 7^2 + 10^2 - 2 \times 7 \times 10 \times \cos 55^\circ$$

$$\therefore AC \approx 8.29$$

(ii)
$$7 \times 10 \times \sin 55^\circ \approx 57.34$$

(e) D: $-3 \leq x \leq 3$
R: $0 \leq y \leq 3$

Question 2 continued

$$(f) \quad (i) \quad m_{AB} = \frac{5-1}{3+2} = \frac{4}{5}$$

$$\therefore y-5 = \frac{4}{5}(x-3)$$

$$\therefore 5y-25 = 4x-12$$

$$\therefore 4x-5y+13=0$$

$$(ii) \quad y+1 = -\frac{5}{4}(x-4)$$

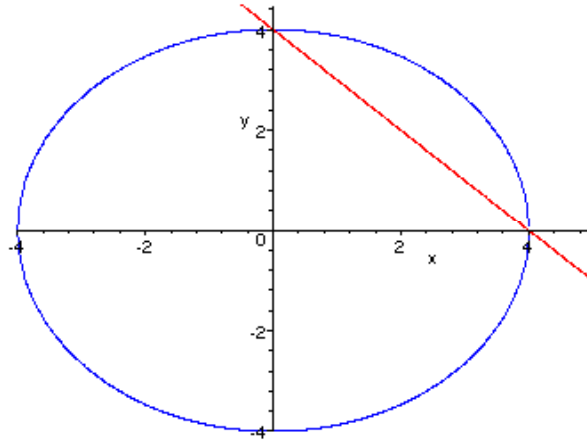
$$\therefore 4y+4 = -5x+20$$

$$\therefore 5x+4y-16=0$$

$$(iii) \quad d_{\perp} = \frac{|Ax_c + By_c + C|}{\sqrt{A^2 + B^2}}$$
$$= \frac{|4 \times 4 + (-5) \times (-1) + 13|}{\sqrt{4^2 + (-5)^2}}$$
$$= \frac{34}{\sqrt{41}} = \frac{34\sqrt{41}}{41}$$

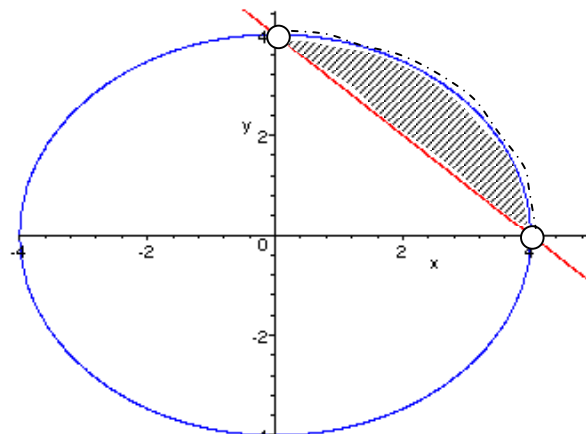
Question 3

(a) (i)



Clearly the points of intersection are $(0,4)$ and $(4,0)$

(ii)



NB the boundary of the circle, in the first quadrant, is dashed and open circles at $(0,4)$ and $(4,0)$.

(b) $\angle BDC = 40^\circ$ (straight line)

$x = 180 - (40 + 37) = 103$ (cointerior angles)

(c) (i) $\angle ADB = \frac{180 - 78}{2} = 51^\circ$ (base angles of isosceles triangle, $AB = BD$)

(ii) $\angle BAC = 180 - 2 \times 78 = 24^\circ$ (base angles of isosceles triangle, $AB = BC$)

$$\begin{aligned}\angle DAC &= \angle BAD - \angle BAC \\ &= \angle ADB - \angle BAC \\ &= 51 - 24 = 17^\circ\end{aligned}$$

Question 3 continued

- (d) (i) Let $m = x^2$
 $\therefore m^2 - 13m - 9 = 0$
 $\therefore m = \frac{13 \pm \sqrt{205}}{2}$
 $\therefore x = \pm \sqrt{\frac{13 \pm \sqrt{205}}{2}}$
- (ii) Let $n = 3^x$
 $\therefore n^2 - 8n - 9 = 0$
 $\therefore (n-9)(n+1) = 0$
 $\therefore n = 9, -1$
 $\therefore 3^x = 9 \quad (3^x \neq -1, \because 3^x > 0)$
 $\therefore x = 2$
- (e) $y^2 - 6y = 2x - 7$
 $\therefore y^2 - 6y + 9 = 2x - 7 + 9$
 $\therefore (y-3)^2 = 2(x+2)$
 NB This is the parabola on its side!
- (i) Axis of symmetry is $y = 3$
- (ii) Vertex is $(-2, 3)$
- (iii) $a = 1/2$
 Focus is $\left(-2 + \frac{1}{2}, 3\right) = \left(-\frac{3}{2}, 3\right)$
 Directrix is $x = -2 - \frac{1}{2} = -\frac{5}{2}$
- (f) $\Delta = [-(k+2)]^2 - 4(3k+6)$
 $= k^2 + 4k + 4 - 12k - 24$
 $= k^2 - 8k - 20$
 $= (k+2)(k-10)$
 Positive definite if $a > 0$ and $\Delta < 0$
 $a = 1, \Delta < 0 \Rightarrow (k+2)(k-10) < 0$
 $\therefore -2 < k < 10$

Question 4

$$(a) \quad (i) \quad \frac{d}{dx} \left(-x^3 + 2x^2 + \frac{1}{2} \right) = -3x^2 + 4x$$

$$(ii) \quad \frac{d}{dx} (\sqrt{7x}) = \frac{d}{dx} ((7x)^{1/2})$$

$$= \frac{1}{2} (7x)^{-1/2} \times 7$$

$$= \frac{7}{2\sqrt{7x}}$$

$$= \frac{\sqrt{7}}{2\sqrt{x}} = \frac{\sqrt{7}}{2} x^{-1/2}$$

$$(iii) \quad \frac{d}{dx} \left(\frac{ax^3 - bx^2 + cx}{x^2} \right) = \frac{d}{dx} (ax - b + cx^{-1})$$

$$= a - cx^{-2}$$

$$= a - \frac{c}{x^2}$$

$$(b) \quad (i) \quad \frac{dy}{dx} = (x+1)^8 \frac{d}{dx} (2x) + 2x \frac{d}{dx} ((x+1)^8)$$

$$= 2(x+1)^8 + 2x \times 8(x+1)^7$$

$$= 2(x+1)^8 + 16x(x+1)^7$$

$$= 2(x+1)^7 [x+1+8x]$$

$$= 2(9x+1)(x+1)^7$$

$$(ii) \quad \frac{dy}{dt} = \frac{(5-2t) \frac{d}{dx} (5+t) - (5+t) \frac{d}{dx} (5-2t)}{(5-2t)^2}$$

$$= \frac{5-2t - (5+t) \times (-2)}{(5-2t)^2}$$

$$= \frac{5-2t+10+2t}{(5-2t)^2}$$

$$= \frac{15}{(5-2t)^2}$$

Question 4 continued

(b) (iii) $f(x) = 3x + x^{-3} \Rightarrow f'(x) = 3 - 3x^{-2} = 3 - \frac{3}{x^2}, f''(x) = 6x^{-3} = \frac{6}{x^3}$

(α) $f'(2) = 3 - \frac{3}{2^2} = 3 - \frac{3}{4} = 2\frac{1}{4}$

(β) $f''(2) = \frac{6}{2^3} = \frac{3}{4}$

(c) (i) Let m be the gradient of the tangent and $m_{\perp}$ the gradient of the normal.

$$y = x^{-2} \Rightarrow \frac{dy}{dx} = -2x^{-3} = -\frac{2}{x^3}$$

$$\therefore m = \frac{dy}{dx}_{x=2} = -\frac{2}{2^3} = -\frac{1}{4}$$

$$\therefore m_{\perp} = 4$$

(ii) $f(-3) + f(4) + f(-1) = [-5] + [4^2] + [2 \times (-1)]$
 $= -5 + 16 - 2 = 9$

(d) (i) $\angle BAD$ is common and $\angle ADB = \angle ABC = 90^\circ$
 so $\triangle ABD \parallel \triangle ACB$ (equiangular).

(ii) $\frac{AB}{AD} = \frac{AC}{AB}$ (all sides are in the same ratio)

$$\therefore \frac{AB}{4} = \frac{9}{AB} \Rightarrow AB^2 = 36$$

$$\therefore AB = 6$$

(e) $\Delta = 16^2 - 4 \times (-4k) = 16(16 + k)$

$$\alpha + \beta = -16, \alpha\beta = -4k$$

(i) $\Delta = 0 \Rightarrow 16(16 + k) = 0$

$$\therefore k = -16$$

(ii) $\Delta > 0 \Rightarrow 16(16 + k) > 0$

$$\therefore k > -16$$

(iii) $\alpha\beta = 1 \Rightarrow -4k = 1$

$$\therefore k = -\frac{1}{4}$$

(iv) $\alpha + \beta = \alpha\beta \Rightarrow -16 = -4k$

$$\therefore k = 4$$

Question 5

(a) (i) $PA = 2PB$.

(ii) $PA = \sqrt{(x+2)^2 + (y-4)^2}, PB = \sqrt{(x-4)^2 + (y-1)^2}$
 $PA = 2PB \Rightarrow PA^2 = 4PB^2$
 $\therefore (x+2)^2 + (y-4)^2 = 4[(x-4)^2 + (y-1)^2]$
 $\therefore x^2 + 4x + 4 + y^2 - 8y + 16 = 4x^2 - 32x + 64 + 4y^2 - 8y + 4$
 $\therefore 3x^2 - 36x + 3y^2 + 48 = 0$
 $\therefore x^2 - 12x + y^2 + 16 = 0$

This is a **circle**.

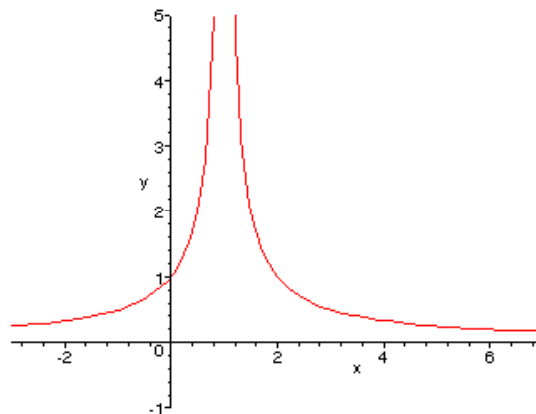
$$x^2 - 12x + 36 + y^2 = -16 + 36$$

$$\therefore (x-6)^2 + y^2 = 20$$

Centre $(6,0)$, radius $2\sqrt{5}$

(b) (i) $x \neq 1$ or $x \in \mathbb{R} \setminus \{1\}$

(ii)
$$y = \begin{cases} \frac{1}{x-1}, & x > 1 \\ \frac{1}{1-x}, & x < 1 \end{cases}$$



Question 5 continued

$$\begin{aligned}
 \text{(c)} \quad & 8 \cos^2 x = 2 \sin x + 7 \\
 & \therefore 8(1 - \sin^2 x) = 2 \sin x + 7 \\
 & \therefore 8 \sin^2 x + 2 \sin x - 1 = 0 \\
 & \therefore \sin x = \frac{-2 \pm \sqrt{36}}{16} = \frac{-2 \pm 6}{16} \\
 & \therefore \sin x = -\frac{1}{2}, \frac{1}{4} \\
 & \therefore x = 14^\circ 29', 165^\circ 31', 210^\circ, 330^\circ
 \end{aligned}$$

$$\begin{aligned}
 \text{(d)} \quad \text{LHS} &= \frac{1}{1 + \sin \theta} + \frac{1}{1 - \sin \theta} \\
 &= \frac{1 - \sin \theta + 1 + \sin \theta}{(1 + \sin \theta)(1 - \sin \theta)} \\
 &= \frac{2}{1 - \sin^2 \theta} \\
 &= \frac{2}{\cos^2 \theta} \\
 &= 2 \sec^2 \theta \\
 &= \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 \text{(e)} \quad & \text{(i)} \quad \tan \frac{\pi}{3} = \frac{AB}{BD} = \frac{AB}{1} \Rightarrow AB = \sqrt{3} \\
 & \text{(ii)} \quad \triangle ABC \text{ is isosceles with } AB = BC. \\
 & \quad DC = BC - BD = \sqrt{3} - 1 \\
 & \text{(iii)} \quad \triangle DEC \text{ is isosceles with } DE = EC. \\
 & \quad 2DE^2 = (\sqrt{3} - 1)^2 \\
 & \quad DE = \frac{\sqrt{3} - 1}{\sqrt{2}} \\
 & \text{(iv)} \quad (\alpha) \quad \angle DAC = \angle BAC - \angle BAD = \frac{\pi}{4} - \frac{\pi}{6} = \frac{\pi}{12} \\
 & \quad (\beta) \quad \sin \angle DAC = \sin \frac{\pi}{12} = \frac{DE}{AD} \\
 & \quad AD^2 = \sqrt{(\sqrt{3})^2 + 1^2} = 2 \\
 & \quad \therefore \sin \frac{\pi}{12} = \frac{\frac{\sqrt{3} - 1}{\sqrt{2}}}{2} = \frac{\sqrt{3} - 1}{2\sqrt{2}}
 \end{aligned}$$

